Answers to Selected Problems

Chapter 1

1. 67,600,000; 19,656,000 **2.** 1296 **4.** 24; 4 **5.** 144; 18 **6.** 2401 **7.** 720; 72; 144; 72 **8.** 120; 1260; 34,650 **9.** 27,720 **10.** 40,320; 10,080; 1152; 2880; 384 **11.** 720; 72; 144 **12.** 24,300,000; 17,100,720 **13.** 190 **14.** 2,598,960 **16.** 42; 94 **17.** 604,800 **18.** 600 **19.** 896; 1000; 910 **20.** 36; 26 **21.** 35 **22.** 18 **23.** 48 **25.** $52!/(13!)^4$ **27.** 27,720 **28.** 65,536; 2520 **29.** 12,600; 945 **30.** 564,480 **31.** 165; 35 **32.** 1287; 14,112 **33.** 220; 572

Chapter 2

9. 74 **10.** .4; .1 **11.** 70; 2 **12.** .5; .32; 149/198 **13.** 20,000; 12,000; 11,000; 68,000; 10,000 **14.** 1.057 **15.** .0020; .4226; .0475; .0211; .00024 **17.** 9.10947×10^{-6} **18.** .048 **19.** 5/18 **20.** .9052 **22.** $(n+1)/2^n$ **23.** 5/12 **25.** .4 **26.** .492929 **28.** .0888; .2477; .1243; .2099 **30.** 1/18; 1/6; 1/2 **31.** 2/9; 1/9 **33.** 70/323 **34.** 1001; 120; 495 **36.** .0045; .0588 **37.** .0833; .5 **38.** 4 **39.** .48 **40.** 1/64; 21/64; 36/64; 6/64 **41.** .5177 **44.** .3; .2; .1 **46.** 5 **48.** .01697 **49.** .4329 **50.** 2.6084×10^{-6} **52.** .09145; .4268 **53.** 12/35 **54.** .0511 **55.** .2198; .0342

Chapter 3

1. 1/3 **2.** 1/6; 1/5; 1/4; 1/3; 1/2; 1 **3.** .339 **5.** 6/91 **6.** 1/2 **7.** 2/3 **8.** 1/2 **9.** 7/11 **10.** .22 **11.** 1/17; 1/33 **12.** .504; .3629 **14.** 35/768; 210/768 **15.** .4848 **16.** .9835 **17.** .0792; .264 **18.** .331; .383; .286; .4862 **19.** 44.29; 41.18 **20.** .4; 1/26 **21.** .496; 3/14; 9/62 **22.** 5/9; 1/6; 5/54 **23.** 4/9; 1/2 **24.** 1/3; 1/2 **26.** 20/21; 40/41 **28.** 3/128; 29/1536 **29.** .0893 **30.** 7/12; 3/5 **33.** .76, 49/76 **34.** 27/31 **35.** .62, 10/19 **36.** 1/2 **37.** 1/3; 1/5; 1 **38.** 12/37 **39.** 46/185 **40.** 3/13; 5/13; 5/52; 15/52 **41.** 43/459 **42.** 1.03 percent; .3046 **43.** 4/9 **45.** 1/11 **48.** 2/3 **50.** .175; 38/165; 17/33 **51.** .65; 56/65; 8/65; 1/65; 14/35; 12/35; 9/35 **52.** .11; 16/89; 12/27; 3/5; 9/25 **55.** 9 **57.** (c) 2/3 **60.** 2/3; 1/3; 3/4 **61.** 1/6; 3/20 **65.** .4375 **69.** 9; 9; 18; 110; 4; 4; 8; 120 all over 128 **70.** 1/9; 1/18 **71.** 38/64; 13/64; 13/64 **73.** 1/16; 1/32; 5/16; 1/4; 31/32 **74.** 9/19 **75.** 3/4, 7/12 **78.** $2p^3(1-p)+2p(1-p)^3$; $p^2/(1-2p+2p^2)$ **79.** .5550 **81.** .9530 **83.** .5; .6; .8 **84.** 9/19; 6/19; 4/19; 7/15; 53/165; 7/33 **87.** 9/16 **90.** 97/142; 15/26; 33/102

Chapter 4

1. $p(4) = 6/91$; $p(2) = 8/91$; $p(1) = 32/91$; $p(0) = 1/91$; $p(-1) = 16/91$; $p(-2) = 28/91$ **4.** (a) 1/2; 5/18; 5/36; 5/84; 5/252; 1/252; 0; 0; 0; 0 **5.** $n-2i$; $i = 0, \ldots, n$ **6.** $p(3) = p(-3) = 1/8$; $p(1) = p(-1) = 3/8$ **12.** $p(4) = 1/16$; $p(3) = 1/8$; $p(2) = 1/16$; $p(0) = 1/2$; $p(-i) = p(i)$; $p(0) = 1$ **13.** $p(0) = .28$; $p(500) = .27$, $p(1000) = .315$; $p(1500) = .09$; $p(2000) = .045$ **14.** $p(0) = 1/2$; $p(1) = 1/6$; $p(2) = 1/12$; $p(3) = 1/20$; $p(4) = 1/5$ **17.** 1/4; 1/6; 1/12; 1/2 **19.** 1/2; 1/10; 1/5; 1/10; 1/10 **20.** .5918; no; $-.108$ **21.** 39.28; 37 **24.** $p = 11/18$; maximum $= 23/72$ **25.** .46, 1.3 **26.** 11/2; 17/5 **27.** $A(p+1/10)$ **28.** 3/5 **31.** p^* **32.** $11 - 10(.9)^{10}$ **33.** 3 **35.** $-.067$; 1.089 **37.** 82.2; 84.5 **39.** 3/8 **40.** 11/243 **42.** 2.8; 1.476 **45.** 3 **50.** 1/10; 1/10 **51.** $e^{-.2}$; $1 - 1.2e^{-.2}$ **53.** $1 - e^{-.6}$; $1 - e^{-219.18}$ **56.** 253 **57.** .5768; .6070 **59.** .3935; .3033; .0902 **60.** .8886 **61.** .4082 **63.** .0821; .2424 **65.** .3935; .2293; .3935 **66.** $2/(2n-1)$; $2/(2n-2)$; e^{-1} **67.** $2/n$; $(2n-3)/(n-1)^2$; e^{-2} **68.** $e^{-10e^{-5}}$ **70.** $p + (1-p)e^{-\lambda t}$ **71.** .1500; .1012 **73.** 5.8125 **74.** 32/243; 4864/6561; 160/729; 160/729 **78.** $18(17)^{n-1}/(35)^n$ **81.** 3/10; 5/6; 75/138 **82.** .3439 **83.** 1.5

Chapter 5

2. $3.5e^{-5/2}$ **3.** no; no **4.** 1/2, .8999 **5.** $1 - (.01)^{1/5}$ **6.** $4, 0, \infty$ **7.** 3/5; 6/5 **8.** 2 **10.** 2/3; 2/3 **11.** 2/5 **13.** 2/3; 1/3 **15.** .7977; .6827; .3695; .9522; .1587 **16.** $(.9938)^{10}$ **17.** .315; .136 **18.** 22.66 **19.** 14.56 **20.** .9994; .75; .977 **22.** .974 **23.** .9253; .1767 **26.** .0606; .0525 **28.** .8363 **29.** .9993 **32.** e^{-1}; $e^{-1/2}$ **34.** e^{-1}; 1/3 **38.** 3/5 **40.** $1/y$

Chapter 6

2. (a) 14/39; 10/39; 10/39; 5/39 (b) 84; 70; 70; 70; 40; 40; 40; 15 all divided by 429 **3.** 15/26; 5/26; 5/26; 1/26 **4.** (a) 64/169; 40/169; 40/169; 25/169; 64/169 **7.** $p(i,j) = p^2(1-p)^{i+j}$ **8.** $c = 1/8$; $E[X] = 0$ **9.** $(12x^2 + 6x)/7$; 15/56; .8625; 5/7; 8/7 **10.** $1/2$; $1 - e^{-a}$ **11.** .1458 **12.** $39.3e^{-5}$ **13.** 1/6; 1/2 **15.** $\pi/4$ **16.** $n(1/2)^{n-1}$ **17.** 1/3 **18.** 7/9 **19.** $-\log(y), 0 < y < 1; 1, 0 < x < 1$; 1/2; 1/4 **21.** 2/5; 2/5 **22.** no; 1/3 **23.** 1/2; 2/3; 1/20; 1/18 **25.** $e^{-1}/i!$ **28.** $\frac{1}{2}e^{-t}$; $1-3e^{-2}$ **29.** .0326 **30.** .3772; .2061 **31.** .0829; .3766 **32.** 5/16; .0228 **33.** e^{-2}; $1 - 3e^{-2}$ **35.** 5/13; 8/13 **36.** 1/6; 5/6; 1/4; 3/4 **41.** $(y+1)^2 xe^{-x(y+1)}$; xe^{-xy}; e^{-x} **42.** $1/2 + 3y/(4x) - y^3/(4x^3)$ **46.** $(1 - 2d/L)^3$ **47.** .79297 **48.** $1 - e^{-5\lambda a}$; $(1 - e^{-\lambda a})^5$ **52.** r/π **53.** r **56.** (a) $u/(v+1)^2$

Chapter 7

1. 52.5/12 **2.** 324; 198.8 **3.** 1/2; 1/4; 0 **4.** 1/6; 1/4; 1/2 **5.** 3/2 **6.** 35 **7.** .9; 4.9; 4.2 **8.** $(1 - (1-p)^N)/p$ **10.** .6; 0 **11.** $2(n-1)p(1-p)$ **12.** $(3n^2 - n)/(4n-2)$, $3n^2/(4n-2)$ **14.** $m/(1-p)$ **15.** 1/2 **18.** 4 **21.** .9301; 87.5755 **22.** 14.7 **23.** 147/110 **26.** $n/(n+1)$; $1/(n+1)$ **29.** $\frac{437}{35}$; 12; 4; $\frac{123}{35}$ **31.** 175/6 **33.** 14, 45 **34.** 20/19; 360/361 **35.** 21.2; 18.929; 49.214 **36.** $-n/36$